

RESEARCH ARTICLE

Enhancing Essay Scoring: An Analytical and Holistic Approach With Few-Shot Transformer-Based Models

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This work was supported by Sungkyul University Research Grant.

ABSTRACT In the field of automated essay scoring (AES), the task of evaluating written compositions has been a persistent challenge. Despite the impressive capabilities of generalized transformer models in various natural language processing (NLP) domains, their application to essay scoring has often fallen short of expectations. In response to this ongoing challenge, this research delves into the intricate nuances of holistic and analytical essay assessment. This work presents an innovative approach centered on Few-Shot transformer-based models, capitalizing on the strengths of pretrained language models while enabling fine-tuning with limited essay-specific data, often called ‘Few-Shot.’ The outcomes of this study are highly promising, with significant improvements in essay scoring accuracy that surpass the performance benchmarks established by conventional methods. The proposed methodology demonstrates remarkable enhancements in the Quadratic Weighted Kappa (QWK) score, indicating its potential. This represents a significant stride towards automating sophisticated essay evaluation, addressing a long-standing issue in the field.

INDEX TERMS AES, NLP, transfer learning, BERT, few-shot learning, holistic scoring, analytical scoring.

I. INTRODUCTION

In the ever-changing world of education and evaluation, grading written essays remains a constant struggle. This difficulty is multifaceted, distinguished by its inherent subjectivity, the requirement of significant resources, and the dependence on expert human opinion. Grading essays requires more than merely subject knowledge; it also requires a thorough comprehension of linguistic nuances, organizational structure, coherence, and persuasive argumentation.

Educators and institutions have long sought to standardize and improve essay-grading processes. However, the introduction of artificial intelligence (AI) and natural language processing (NLP) technologies has ushered in a new era of innovation. These developments can significantly

change AES, providing greater efficiency, consistency, and impartiality in the evaluation process.

At the heart of this transformation are transformer-based models, which hold the potential to revolutionize AES. These models offer the tantalizing promise of making scoring faster, more consistent, and unbiased. AES has attracted the interest of researchers across various fields, including linguistics, education [1], [2], [3], and NLP [4], [5].

Currently, AES models are divided into two categories: traditional approaches that rely on manually crafted features and neural network-based approaches that process raw text input. The traditional approach includes systems like Project Essay Grade (PEG) [6], which evaluates essays using features such as word count, grammar checks, and n-gram representations. Another example is the use of Latent Semantic Analysis (LSA) [7], which applies statistical methods to assess the semantic similarity of essay content. However, these methods

The associate editor coordinating the review of this manuscript and approving it for publication was Wai-Keung Fung.

face constraints such as the need for custom-tailored features and limited adaptability across diverse evaluation tasks. In contrast, neural network-based approaches have emerged as promising alternatives by eliminating the dependency on manual feature engineering. For example, models like Recurrent Neural Networks (RNNs) [8], [9] demonstrate the ability to analyze sentence coherence and argument strength effectively. However, these deep neural networks (DNNs) require large amounts of labeled data, which is not always feasible in practical scenarios. This limitation presents a significant challenge in domains such as AES, where acquiring large-scale labeled datasets for every essay prompt or genre can be impractical.

A promising solution to this problem is Few-Shot learning, a machine learning paradigm that enables models to generalize effectively from limited annotated examples. Unlike traditional approaches, which rely heavily on manually crafted features, Few-Shot learning eliminates the need for extensive feature engineering by leveraging pretrained models. Furthermore, it surpasses neural-based approaches, which typically depend on deep neural networks requiring large-scale labeled datasets for training. Few-Shot learning's ability to adapt quickly to new tasks with minimal data makes it particularly advantageous in domains like Automated Essay Scoring (AES), where acquiring labeled data for every unique essay prompt or genre is impractical. This paradigm bridges the gap between the scalability of neural methods and the interpretability of traditional approaches, offering an efficient and flexible solution for diverse educational contexts.

Another exciting avenue of research lies in the potential of pretrained models to be fine-tuned across a wide range of tasks, reducing the need to develop new models from scratch. Among these pretrained models, Bidirectional Encoder Representations from Transformers (BERT) [10] is regarded as the apex of pretrained language models. Based on a multi-layer bidirectional transformer architecture, BERT has exhibited impressive performance across various language-related tasks. Despite its success in other fields, BERT's application in AES has yet to be discovered, allowing plenty of space for research and innovation.

In the broader context of education, it has been asserted that nothing is more critical than assessing students' work and providing feedback. Both scores and feedback are fundamental to the learning process. Accurate scoring ensures fairness in the evaluation of student work and forms the foundation of their learning experience. Simultaneously, offering comprehensive feedback on their responses is a guiding compass, helping students identify weaknesses and enhancing performance.

Rubrics emerge as indispensable tools within the intricate process of evaluating students' responses to essay prompts. These rubrics consist of clear criteria and performance level descriptions for each criterion, ensuring consistency and fairness in grading [11]. Rubrics can be broadly categorized into two principal types: analytical and holistic rubrics.

Analytical rubrics, or trait-specific scoring, evaluate each criterion independently, providing detailed feedback aligned with instructional objectives. For instance, [12] employed transfer learning and DNN to score multi-dimensional essays, while [13] proposed an RNN-based system to predict multiple trait scores simultaneously. These approaches enhance scoring precision but demand significant time and inter-rater reliability efforts.

In contrast, AES systems have traditionally used holistic scoring, assigning a single score based on the essay's overall impression. Reference [14] introduced a system using coherence and self-attention to score essays based on grammar and structure, and [15] suggested a self-supervised cross-prompt holistic scoring approach. While efficient, holistic scoring has been criticized for lacking detailed feedback. Hybrid methods combining analytical and holistic scoring are emerging to address these limitations, enabling targeted improvements in writing quality.

Research shows that learners prefer AES systems feedback over peer feedback [16]. AES systems have made significant advancements, autonomously assessing written content and assigning scores. Deep learning and neural network models facilitate this transformation. However, most existing neural-based AES systems focus on holistic scoring, neglecting the assessment of traits [17], which are essential to delineate learners' performance levels, especially in cross-prompt assessment.

A key challenge in developing robust AES systems lies in the scarcity of annotated datasets, especially for new or less common essay types. Few-Shot learning, a paradigm designed to generalize from minimal labeled examples, offers a transformative solution. Using the power of pretrained models, Few-Shot learning enables AES systems to perform effectively across diverse prompts and genres, reducing dependence on large-scale task-specific datasets. This capability is particularly significant in educational contexts, where acquiring labeled essays for every unique evaluation scenario is impractical.

As the landscape of education and assessment continues to evolve, there is a growing need for an effective AES system that seamlessly integrates transformative components, such as transformer-based models, trait-based scoring, and comprehensive feedback even with limited data available. This research embarks on a journey to explore, innovate, and redefine the AES paradigm, unveiling new possibilities and elevating the standards of essay assessment.

In this study, we selected BERT as the model for automated essay scoring due to its exceptional performance across various natural language processing tasks, including text classification and sentiment analysis. BERT's bidirectional architecture allows it to capture contextual relationships between words effectively, making it particularly well-suited for evaluating the complexity, coherence, and structure of written essays, which are essential in AES tasks. Compared to other LLMs, BERT offers a balanced trade-off between

accuracy and computational efficiency, making it a more practical choice for resource-constrained environments. Furthermore, its pre-trained nature enables fine-tuning with minimal labeled data, aligning perfectly with our Few-Shot learning setup while ensuring robust generalization across diverse prompts and scoring criteria.

The potential impact of our proposed method, which fine-tunes the BERT base model and implements Few-Shot Training, on the field of AES is significant, as it can revolutionize the way essays are evaluated, making the process more efficient, accurate, and adaptable to diverse essay topics.

The contributions of this paper include:

- Introduces a groundbreaking method by fine-tuning the BERT base model, allowing automated assessment from diverse perspectives without manual intervention.
- Implementing Few-Shot Training with pretrained BERT, the proposed approach significantly enhances the reliability and accuracy of the scoring model. This is validated through rigorous experiments on the ASAP dataset, where the proposed method achieves outstanding results in both holistic and analytical scoring, establishing a new benchmark for effectiveness in automated essay evaluation.
- Advances the performance of AES models on diverse essay topics, contributing to making essay evaluation more efficient and accurate across various aspects.

A. PROBLEM STATEMENT

Traditional manual scoring encounters challenges like subjectivity, inconsistency, and resource intensity, hindering scalability and reliability, especially with extensive datasets. Transformer-based models have garnered attention, but their use in AES faces hurdles. Limited generalizability raises concerns about their effectiveness in diverse educational contexts. Additionally, the lack of domain-specific knowledge in these models poses several challenges, given the diverse subjects in essays, requiring detailed understanding to assign scores based on any rubric. A specialized AES transformer-based model that is fine-tuned for educational essay scoring is urgently needed.

The layout of this paper is as follows. Section II of the paper presents a comprehensive review of the existing literature on essay scoring methodologies. Section III presents a detailed explanation of the proposed method, including the complete framework developed. Section IV describes the setup, and section V primarily focuses on evaluating the proposed model. Finally, VI and VII are presented in the last sections of the paper.

II. LITERATURE REVIEW

Upon meticulous examination of each article, the methodologies employed by automated essay grading systems have been categorized into two distinct groups: conventional and neural network strategies. Furthermore, identification and

TABLE 1. Notations used within the paper.

Annotation	Expanded Form
AES	Automated Essay Scoring
DL	Deep Learning
NLP	Natural Language Processing
BERT	Bidirectional Encoder Representations from Transformers
ERNIE	Enhanced Representation via kNowledge Integration
ASAP	Automated Student Assessment Prize
GELU	Gaussian Error Linear Units
QWK	Quadratic Weighted Kappa
MSE	Mean Squared Error

classification into four types have been conducted within the neural network strategies, as delineated by [18] in the study.

A. TRADITIONAL APPROACH USING HANDCRAFTED FEATURES

Traditional automated essay grading systems evaluate and grade works based on precisely hand-created features. The efficiency of such systems is highly dependent on the quality of the picked attributes. Despite its effectiveness, manually building the features is a lengthy process due to the vast amount of available data.

Reference [6], a former high school English teacher, devised this automated scoring method in the 1960s after “many long weekends going through stacks of papers” and asking for assistance from colleagues. (Project Essay Grade) PEG was adopted in the mid-1980s due to computers’ low automation capabilities at the time of Page’s suggestion. The 1990s revitalized PEG. Reference [7] suggested a statistical model for automatic essay grading. They used different features such as character count, misspelled words, and n-gram representation to create an essay vector. Latent Semantic Analysis (LSA) was applied to a matrix of vectors to score each article.

The Timed Aggregate Perceptron vector model, introduced by [19], scores essays based on a combination of variables. The model was trained using word character count, misspelled words, and n-gram representation. The rank of an essay was used to estimate the essay grade.

Reference [20] put forth a text-mining approach for brief answer evaluation. They contrasted the model’s and students’ responses by measuring the distance between the sentences. Reference [21] proposed using Mutual Information Regression to show the impact of feature selection. Four ML methods were examined using linguistic features. The 3-layer network with feature selection outperformed all ML methods. A new hybrid model integrated selected traits with a higher-level deep neural network to gain deep learning benefits. (DNN). The suggested hybrid model was more accurate.

B. NEURAL NETWORKS APPROACH

This study utilizes DL techniques within the neural network paradigm to explore and elaborate on four approaches to AES.

1) PROMPT-SPECIFIC HOLISTIC SCORING

In an early DNN-AES model by [8], essays are assessed using recurrent neural networks (RNNs). Each essay word is initially transformed into G-dimensional word embeddings via a Lookup table layer. CNNs capture local textual associations in the convolutional layer to create word embeddings. The recurrent layer employs an LSTM network to identify time series dependencies and generates a vector at each timestep.

A deep neural network-AES model with a three-layer RNN architecture was proposed by [9]. For the part of the recurrent layer, the model uses the last pooling and a bidirectional LSTM. The mentioned framework is distinguished by the use of Score-Specific Word Embedding (SSWE), an evolution of word embeddings by [22]. In terms of word embedding essay quality, SSWE exceeds C&W.

Learning vector quantization with a neural network was presented by [23] for essay training. Training lets the network score ungraded essays. After preprocessing the framework, the relevance of the essay will be evaluated.

Reference [24] proposed a study that introduces a method combining RoBERTa embeddings with linguistic features in Lightweight XGBoost for improved scoring. RoBERTa captures essay context, while handcrafted features address grammar and readability.

The previous models interpret an essay as words. Nevertheless, [25] advocated modeling text hierarchies. Word and phrase CNNs were created for a two-level hierarchical representation technique. A hierarchical CNN model for automated essay grading showed that automatically learned features were competitive with handcrafted features for both in-domain and domain-adaptation tasks.

Reference [26] proposed an attention mechanism, initially proposed by [27], for a hierarchical representation model focusing on specific input data areas that allow the attention mechanism neural network design to make predictions accurately.

Reference [14] proposed an AES system using coherence and self-attention. The essay score is calculated from the coherence score and other factors like grammar. Based on their strengths, [28] advocated that traditional feature engineering and neural network methods should be used together rather than in opposition.

Reference [29] put forth a hybrid technique to take advantage of both methods. They propose a hybrid method incorporating hand-created essay-level characteristics into a Deep Neural Network-AES model.

Reference [30] suggested an essay evaluation sequence-to-sequence learning model. BERT and XLNet were both employed for essay grading. Reference [4] compared AES models based on manually extracted features, word2vec representation, and BERT model word embedding.

Reference [31] present a novel essay scoring method to simultaneously learn a multi-scale essay representation with BERT, significantly improving the outcome compared to the conventional method of employing pretrained language models.

2) PROMPT-SPECIFIC ANALYTICAL SCORING

A DNN-AES model was used by [32] to rate prompt-specific features. Trait-specific essay outcomes were predicted using a hierarchical representational model with an attention mechanism. Reference [13] proposed a system that specializes in scoring prompt-specific trait jobs and can simultaneously predict multiple trait scores. Built on the RNN-based design, this model is intended to have several outputs.

3) CROSS-PROMPT HOLISTIC SCORING

Reference [33] using scored essays for non-target prompts, the two-stage deep neural network (TDNN) method trains a scoring model independent of the prompt and creates pseudo-rating data for unrated target prompt essays.

According to [34], the SEDNN framework, which stands for Shared and Enhanced Deep Neural Network, would extract more compelling features. This study offered a two-stage paradigm for an automated cross-prompt essay assessment.

Reference [15] suggested a holistic scoring approach for cross-prompt for the automatic essay scoring job employing two distinct learning tasks that are self-supervised for each prompt. Reference [12] suggested employing transfer learning and DNN to score multi-dimensional essays. A hierarchical fine-tuning strategy is utilized to fine-tune a BERT framework for essay grading.

4) CROSS-PROMPT ANALYTICAL SCORING

Reference [35] proposed predicting essay coherence and argument strength using RNN-based models. Reference [36] developed a trait scoring model that predicts multiple trait scores. Cross-prompt Trait Scorer (CTS)1 forecasts holistic scoring and all trait scoring simultaneously. Reference [12] suggested employing transfer learning and DNN to score multi-dimensional essays. They provided results on the CELA dataset.

In the realm of PAES, [37] introduced a domain-generalized paradigm rooted in Multi-Task Learning. Their approach was to create a versatile essay scoring system adept at effectively handling various prompts.

Additionally, [38] put forth the ProTACT model, which emphasized cross-prompt essay trait scoring. This innovative architectural design seamlessly integrated prompt-specific and trait-specific encodings, augmenting the model's capacity to discern nuanced essay traits across a spectrum of prompts while retaining prompt-specific characteristics.

C. LLM APPROACH

This review includes studies on ChatGPT's role in educational assessment to address recent advancements in AES, particularly with large language models (LLMs). Recent research highlights the utility of LLMs in standardized testing and evaluation scenarios, showcasing ChatGPT's potential to complement or enhance traditional assessment methods. For instance, ChatGPT has demonstrated capabilities

in generating explanations for scoring through rationale generation, enhancing interpretability in student answer assessment [39]. ChatGPT’s performance on exams such as the GRE and Vietnamese National High School Graduation Examination also illustrates its problem-solving competencies and adaptability to various academic contexts [40], [41].

Alongside, [42] introduce a framework leveraging LLMs for zero-shot essay scoring by decomposing writing proficiency into distinct traits and generating specific scoring criteria for each. The study uses Llama2-13b-chat to assess essays effectively without prior task-specific training. Reference [43] investigate the collaboration between human graders and LLMs, including proprietary and open-source models, in essay scoring. The study reveals that while LLMs may not outperform state-of-the-art grading models, they offer notable consistency, generalizability, and explainability, enhancing the grading process alongside human evaluators.

Evaluations of newer multimodal LLMs like ChatGPT-4 Vision on Brazil’s National Undergraduate Computer Science Exam further underscore advancements in using LLMs for educational purposes, mainly where multimodal assessment is crucial [44]. These developments highlight the transformative impact of ChatGPT and similar models on educational assessment, emphasizing the increasing relevance of LLM-based methodologies in AES.

Reference [45] also responded to other concerns regarding this research, including datasets, features, assessment matrices, and various techniques.

Examining available literature underscores the significance of AI technology such as AES for educators, particularly in the wake of recent pandemic conditions [46]. The importance extends across all four types of AES in the contemporary era. Studies indicate that systems offering analytic scoring and detailed feedback on written essays garner increased student trust [47].

Table 2 explains the AES approaches, describing the type, model, and dataset used.

Figure 1 displays the number of publications per year for the papers cited in our study. This chart highlights trends in research output among the works referenced, providing insights into the temporal distribution of impactful contributions relevant to our research. Figure 2 illustrates the proportional contributions of the different approaches employed for AES that are referenced in this study.

D. FEW-SHOT LEARNING APPROACH

Recent NLP advancements have showcased the potential of Few-Shot learning frameworks, such as zero-shot and Few-Shot setups, for various text classification and generation tasks. Studies by [42] demonstrated the effectiveness of Few-Shot learning for decomposing writing proficiency into distinct traits and generating specific essay scoring criteria. Similarly, [43] explored Few-Shot methods for combining human feedback and model-generated predictions in educational assessments, achieving robust generalization

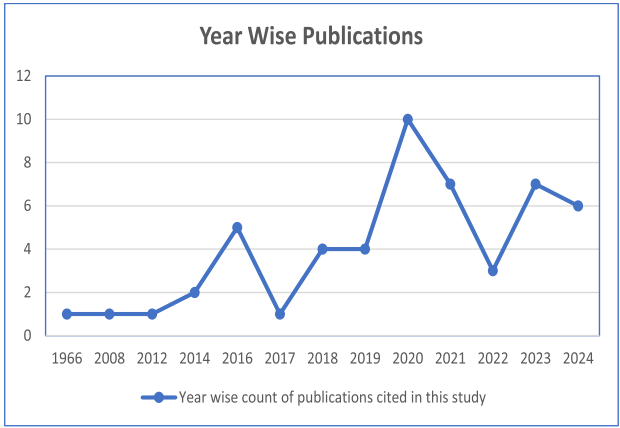


FIGURE 1. Year wise publications of papers cited in this study.

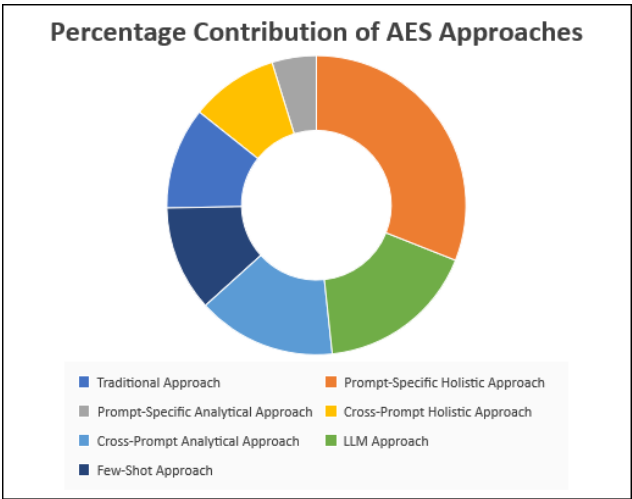


FIGURE 2. Percentage contribution of approaches for AES referenced in this study.

capabilities. Reference [48] introduced the Transductive Graph-based Ordinal Distillation (TGOD) framework under the one-shot AES paradigm, leveraging Few-Shot techniques to generate pseudo-labels from a single labeled essay per score. Reference [49] investigated the use of Few-Shot learning for Automated Essay Scoring (AES) by applying pattern-exploiting training (PET) and SetFit techniques. Their findings showed significant performance improvements with PET, especially in situations with limited data, although it proved to be computationally demanding. In contrast, while SetFit was faster, it showed reduced effectiveness in certain scenarios. These methods underscore the ability of Few-Shot learning to adapt pretrained language models like BERT and GPT to specialized domains with minimal additional training data.

Compared to traditional AES methods, which rely heavily on handcrafted features and rule-based scoring systems, Few-Shot learning offers a flexible and scalable alternative. Unlike neural approaches that typically require large-scale

TABLE 2. Overview of the approaches used for AES.

Types of AES	Models	Authors	Dataset
Traditional Strategy	Machine Learning	[6]	N/A
		[19]	ASAP
		[20]	The Introductory Comp Sc. (Short Qs)
		[21]	ASAP
Prompt Specific Holistic Scoring	Frameworks Based on RNN	[8]	ASAP
	Hierarchical Representation Frameworks	[9]	ASAP
		[26]	ASAP
	Hybrid Framework	[14]	ASAP & CEEE
		[28]	ASAP
	BERT Based Frameworks	[30]	ASAP
		[4]	ASAP
Prompt Specific Analytical Scoring	Multiple Models for Specific Trait Framework	[31]	ASAP
		[32]	ASAP & ASAP++
		[13]	ASAP
Cross Prompt Holistic Scoring	Two Stage Learning Framework	[33]	ASAP
	Self Supervised Learning Framework	[15]	ASAP
	Multi-Dimensional Transfer Model	[12]	ASAP & CELA
Cross Prompt Analytical Scoring	Multiple Models for Specific Trait with Self-Supervised Learning Framework	[35]	ICLE(International Corpus of Learner English)
	Multiple Outputs Framework	[36]	ASAP
	Multi-Dimensional Transfer Model	[12]	ASAP & CELA

datasets for training, Few-Shot learning mitigates the need for extensive labeled data by transferring knowledge from pretrained models to specific AES tasks. This approach reduces computational overhead and enables prompt-specific generalization, making it a promising solution for real-world AES applications.

III. METHODOLOGY

In this study, we selected BERT as the core model for Automated Essay Scoring (AES) due to its proven capabilities in capturing contextual relationships in text and achieving state-of-the-art results across a range of natural language processing tasks. BERT's pre-trained nature, which allows for fine-tuning with minimal labeled data, makes it especially well-suited for our Few-Shot Learning approach. This capability is critical in scenarios where labeled data is scarce, such as essay scoring, where obtaining large annotated datasets for every essay type is often impractical. Moreover, BERT's bidirectional attention mechanism enhances its ability to understand both the syntactic and semantic nuances of essays, which are essential for scoring tasks.

The methodology encompasses key components collectively enhancing AES using BERT-based models, as depicted in Figure 3. It initiates with a robust data preprocessing phase, exploring the BERT model, including its input and weight-sharing modules. Two variants of BERT, namely BERT Base Cased and BERT Base Uncased, are investigated to highlight their performances. Subsequently, the Few-Shot Training

Module is explored as a critical element for adapting the model with minimal annotated examples. Finally, the Output Module is presented, operating differently for holistic and analytical scoring, utilizing Single-Task Learning for the former and Multi-Task Learning for the latter. These components collectively constitute the comprehensive methodology for practical and versatile essay scoring. Figure 3 illustrates an overview of the proposed approach, showcasing the integration of the BERT model with Few-Shot Training.

A. PREPROCESSING DATA (INPUT MODULE)

Data preparation is used in BERT-based AES research to turn raw essay data into a structured format for analysis. This procedure includes lowercasing, punctuation removal, tokenization, stopword removal, and stemming. Lowercasing maintains uniformity and eliminates case sensitivity difficulties while removing punctuation refines information for more accurate judgments. Tokenization divides text into separate words, whereas stemming lowers words to their root forms, improving semantic similarity recognition and the general resilience and accuracy of the AES system. Figure 4 depicts the input module that does essay preprocessing.

B. BERT MODEL MODULE

The BERT model plays a central role in revolutionizing AES in this research. BERT's bidirectional architecture empowers it to understand language context, making it ideal for essay grading. It processes essays as sequences, capturing rich

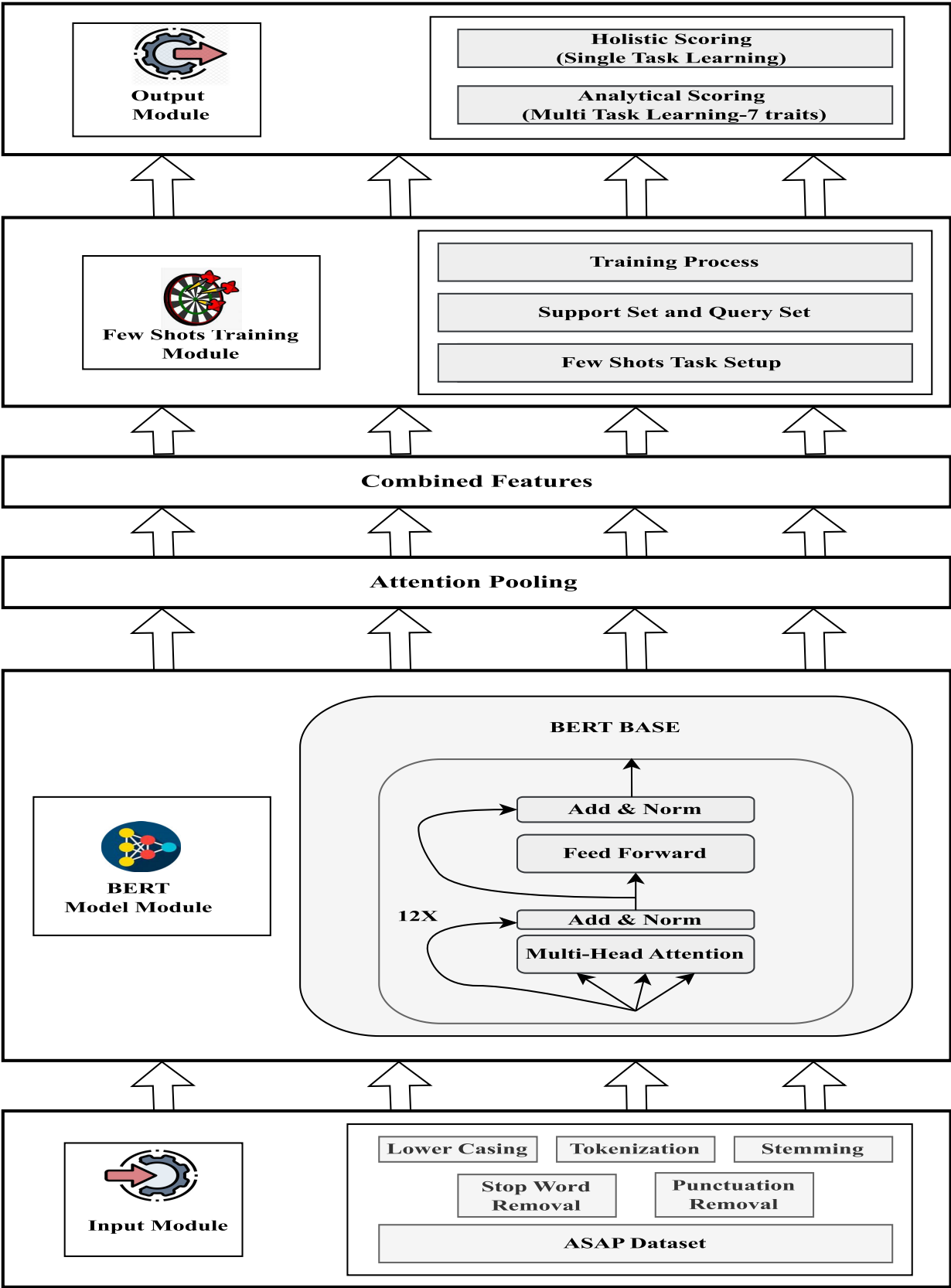


FIGURE 3. Proposed model.

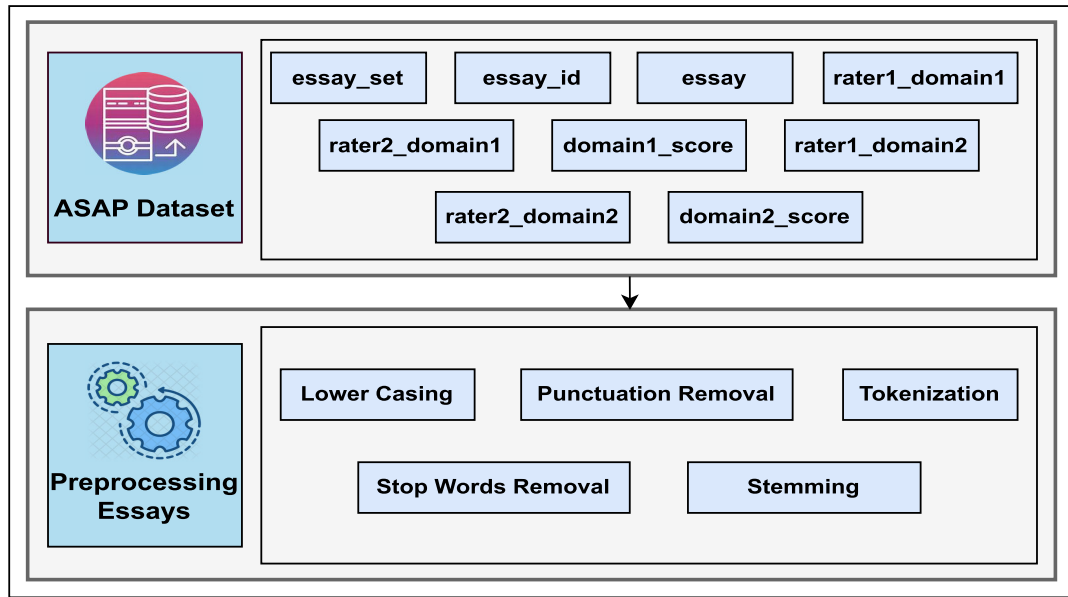


FIGURE 4. Input module.

contextual information often missed by traditional AES models. Figure 5 gives the overview of the BERT module.

1) INPUT MODULE OF BERT

In the input module, there are two steps: tokenization and embedding. The input representation module tokenizes the essay text into subwords or tokens $Token_1, Token_2, \dots, Token_n$. These tokens are then transformed into high-dimensional embeddings using the embedding matrix E . This process can be represented as in Equation 1:

$$E(Token_i) [for i = 1, 2, \dots, n] \quad (1)$$

Here, $E(Token_i)$ signifies the embedding of each token, preparing the input for subsequent layers of the BERT model.

2) WEIGHT SHARING MODULE WITH ATTENTION POOLING

Before exploring the specifics of self-attention and weighted sum computations, it is vital to understand how BERT uses such processes to capture contextual information and build meaningful representations for AES. The core of BERT lies in its weight-sharing mechanism and use of self-attention. BERT uses a multi-layer bidirectional Transformer architecture. Each layer includes self-attention techniques and feedforward neural networks.

a: SELF-ATTENTION MECHANISM

In the context of AES, the BERT model consists of 12 encoders, each with 12 bidirectional self-attention heads and 768 hidden units. In the weight-sharing component, these encoders use a self-attention method to calculate scores for each token in the embedding vectors. BERT employs self-attention to calculate attention weights. The calculation

of attention weights, $AttentionWeight$, can be expressed as Equation 2 for a specific token, $Token_i$, in the essay:

$$AttentionScores_i = \text{softmax}(Q_i K_i^T) \quad (2)$$

Here, Q_i represents the query vector for $Token_i$, and K_i represents the key vectors for all tokens in the essay. The softmax function normalizes the attention scores.

b: WEIGHTED SUM OF TOKENS

The weighted sum of token embeddings using attention weights can be represented as in Equation 3:

$$Token'_i = \sum_{j=1}^n AttentionWeight_{i,j} \cdot E(Token_j) \quad (3)$$

The equation provided calculates $Token'_i$, the final representation of token ' i ' in the sequence. It takes a weighted sum of the embeddings of all other tokens ' j ' in the sequence. These weights ($AttentionWeight_{i,j}$) determine how much attention token ' i ' gives to each token ' j ' based on their contextual relevance. This process enables token ' i ' to gather information from the entire sequence, incorporating the importance of each token as determined by the attention mechanism.

c: WEIGHT SHARING

Each layer in BERT has its own set of parameters for self-attention and feedforward neural networks. The self-attention mechanism in each layer utilizes distinct weight matrices (W_Q, W_K, W_V , etc.) to compute attention scores and outputs.

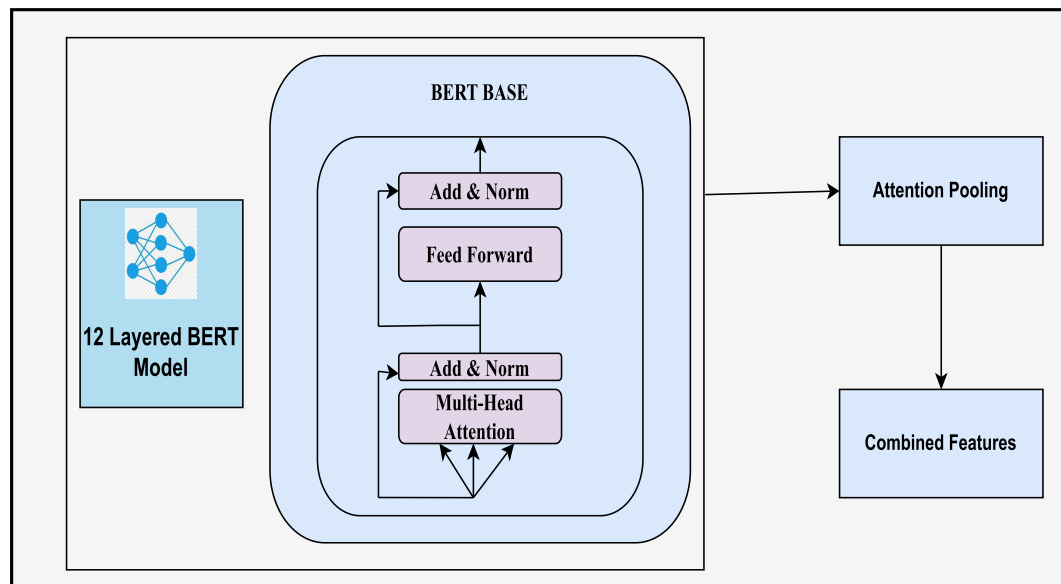


FIGURE 5. BERT model module.

C. FEW-SHOT TRAINING MODULE

The Few-Shot Training module is a pivotal component of the proposed essay-scoring approach. With limited annotated examples in the support set, this BERT-based model learns to assess essays in the query set accurately. This adaptability makes the proposed system versatile and efficient, reducing the need for extensive labeled data and enhancing model generalization. Few-Shot learning significantly contributes to the accuracy and robustness of the proposed AES methodology. Figure 6 demonstrates the Few-Shot Training Module.

In addition to holistic scoring, Few-Shot Training is equally valuable for analytical scoring, which involves assessing specific linguistic aspects of essays. Here, Few-Shot learning enables the proposed model to focus on particular linguistic nuances, coherence, and organization, adapting to the exact analytical criteria with a minimal number of examples. This versatility of the proposed system allows it to cater to a wide range of essay evaluation needs. Here is how Few-Shot Training works:

1) CONSTRUCTING FEW-SHOT TASKS

In the context of this research on AES using the ASAP dataset, constructing Few-Shot tasks is a crucial step that empowers the proposed model to make accurate essay evaluations. Few-Shot learning entails providing the proposed transformer-based model, BERT, with limited annotated examples, allowing it to generalize effectively and assess essays it has not encountered before.

a: SUPPORT SET

The support set is a fundamental element of Few-Shot Training. It comprises a small number of essays, and in this

proposed approach case, 120 essays for each of the eight prompts are present in the ASAP dataset. These essays serve as the context and foundation for the model to understand the traits, linguistic features, and nuances relevant to each prompt. The support set guides the model on what to look for when assessing essays.

The choice of 120 essays from each prompt stems from the need for sufficient yet manageable contextual information. It strikes a balance between providing the model with valuable examples to learn from and ensuring that the support set remains manageable.

b: QUERY SET

The query set, distinct from the support set, encompasses essays not included in the support set. Specifically, 80 essays were chosen for each of the eight prompts in the ASAP dataset to form the query set. Its main objective is to assess the model's capacity for generalization based on the information acquired from the support set.

The selection of 80 essays in the query set considers the need for diverse evaluations. It ensures that the model can accurately score essays it has yet to be explicitly trained on, thereby validating the proposed AES system's robustness and generalization capabilities.

In designing the Few-Shot learning approach, the selection of 120 essays for the support set and 80 essays for the query set was based on a balance between providing sufficient examples for effective model adaptation and maintaining computational efficiency. The support set size of 120 essays ensures a diverse representation of linguistic traits across various prompts, which is crucial for the model to generalize effectively to new prompts. Meanwhile, a query set of 80 essays offers a robust sample for evaluating model

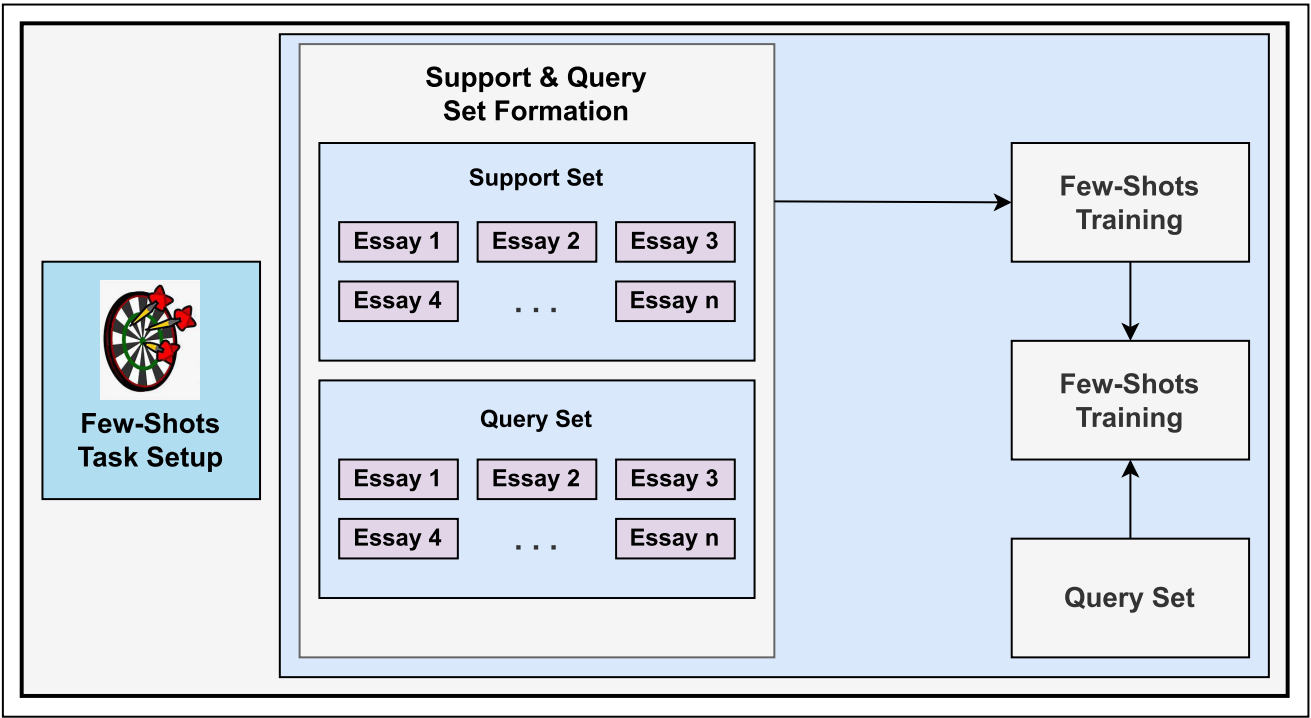


FIGURE 6. Few-shot module.

performance, allowing us to assess the model’s ability to generalize beyond the examples seen during training. This balance minimizes overfitting risks while ensuring that the model receives ample data to learn nuanced scoring patterns within the constraints of Few-Shot learning.

In summary, constructing Few-Shot tasks employs a systematic approach to furnish the BERT-based model with context and examples necessary for accurate essay assessment. The support set serves as a foundation for learning specific prompts, while the query set ensures the model’s generalization and evaluation capabilities extend beyond the support set. This methodology, tailored to the nuances of the ASAP dataset, achieves accurate and versatile AES.

2) TRAINING

To give reliable essay evaluations, the AES system employs BERT-based Few-Shot learning. The method starts with creating support and query sets and then moves to the training phase. The BERT-based model is fine-tuned using information from the support set essays. This iterative procedure assists the model in comprehending essay quality and aligning its comprehension with each essay challenge. The model grows over time, improving its versatility and reflecting the complexities of various articles. The knowledge gained during the training phase is strategically leveraged in the model prediction stage, bridging the gap between generalized models and essay scoring complexity.

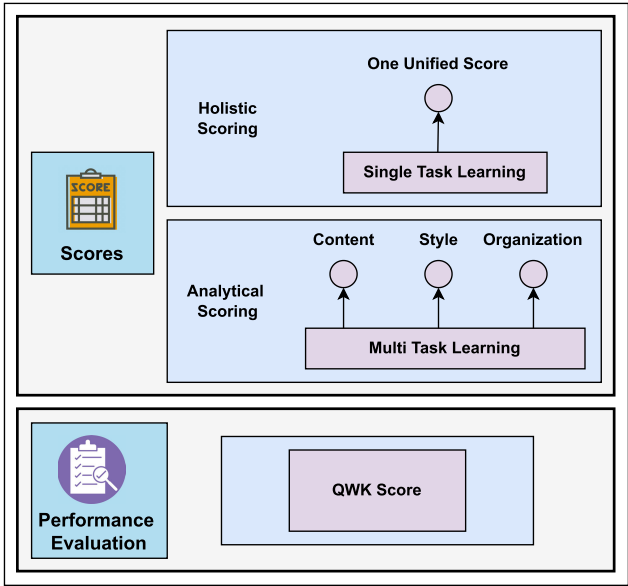


FIGURE 7. Output module.

3) PREDICTION

The prediction module is a component of an AES system that employs Few-Shot BERT training. It evaluates essays from the query set using a trained model, exhibiting its generalization ability. The model then processes each essay using the qualities it learned from the training set (support

set). It assigns scores based on holistic and analytical evaluation criteria. The model analyses coherence, content, organization, and language use for a complete study and predicts each attribute. This sophisticated method bridges the gap between generalized models and essay scoring, resulting in an accurate and efficient AES system.

D. OUTPUT MODULE

The output module in AES using BERT-based Few-Shot learning is the final step in the model's architecture, producing predictions for essay scores. This module operates differently for holistic and analytical scoring. Figure 7 represents the output module.

1) HOLISTIC SCORING (SINGLE-TASK LEARNING)

After processing the essay through the BERT model and fine-tuning it with Few-Shot Training, the features are aggregated to form a holistic essay representation. This representation captures the essay's global quality and coherence. This score is a holistic essay assessment, providing a numerical measure of its overall quality. This score represents the model's judgment of how well the essay addresses the prompt and maintains coherence.

Algorithm 1 demonstrates the holistic scoring procedure:

Algorithm 1 Holistic Scoring Procedure

Input: D (essays, holistic scores), BERT model, Hyperparameters.

Initialization: $\theta \leftarrow$ Initialize model parameters, Copy BERT layers, Random init task layers.

Mini-batches: Merge rating dimensions in D .

Training:

- epoch $\leftarrow 0$.
- While epoch $< \text{max_epochs}$:
 - Shuffle mini-batches D .
 - Increment the epoch counter.
 - For each mini-batch $batch_i$ in D :
 - * Calculate the loss for holistic score and its gradient $\nabla(\theta)$.
 - * Update the model parameters using the gradient descent step: $\theta \leftarrow \theta - \eta \nabla(\theta)$, where η is the learning rate.

Few-Shot Training:

- For each episode:
 - Sample a task T .
 - Select support set S_T and query set Q_T from D .
 - Fine-tune the model on S_T using AdamW.
 - Test the model on Q_T for evaluating analytical scores.
-

2) ANALYTICAL SCORING (MULTI-TASK LEARNING)

Analytical scoring, on the other hand, involves assessing specific traits or aspects of an essay, such as content,

organization, vocabulary, and grammar. Analytical scoring is often also referred as trait-specific scoring. Analytical scoring often employs Multi-Task Learning, where the model simultaneously predicts scores for all traits. Multi-Task Learning is a technique that allows the model to predict multiple outputs at the same time, each corresponding to a different aspect of the essay. In the context of analytical scoring, the output module consists of separate branches, each dedicated to evaluating a distinct trait, such as content, organization, and word choice. These branches independently assess specific aspects of the essay's quality, enabling a more fine-grained evaluation of various traits. By training the model to optimize predictions for all traits concurrently, Multi-Task Learning improves the model's ability to handle complex, multi-dimensional evaluations of essays. Algorithm 2 demonstrates the procedure for analytical scoring.

Algorithm 2 Analytical Scoring Procedure

Input: D (essays, holistic scores), BERT model, Hyperparameters.

Initialization: Initialize model parameters θ , Copy BERT layers and randomly initialize task-specific layers for analytical scoring (one for each trait).

Mini-batches: Merge rating dimensions in D .

Training:

- epoch $\leftarrow 0$.
- While epoch $< \text{max_epochs}$:
 - Shuffle mini-batches D .
 - Increment the epoch counter.
 - For each mini-batch $batch_i$ in D :
 - * Calculate the loss for analytical scores and its gradient $\nabla(\theta)$.
 - * Update the model parameters using the gradient descent step: $\theta \leftarrow \theta - \eta \nabla(\theta)$, where η is the learning rate.

Few-Shot Training:

- For each episode:
 - Sample a task T .
 - Select support set S_T and query set Q_T from D .
 - Fine-tune the model on S_T using AdamW.
 - Test the model on Q_T for evaluating analytical scores (Different Trait Scores).
-

IV. EXPERIMENTAL SETUP

The Following Sections gives the overview of the experimental setup of proposed approach.

A. DATASET

The 2012 Automated Student Assessment Prize (ASAP) dataset,¹ a significant contribution to AES research, is the result of a fruitful collaboration between Kaggle and the Hewlett Foundation. This dataset has played a central role in

¹<https://www.kaggle.com/c/asap-aes>

TABLE 3. Summarization of details of ASAP dataset.

Prompt ID	Essay Type	No. of Essays	Avg. Length
1	Argumentative / Persuasive	1785	350
2	Argumentative / Persuasive	1800	350
3	Source-Dependent Response	1726	150
4	Source-Dependent Response	1772	150
5	Source-Dependent Response	1805	150
6	Source-Dependent Response	1800	150
7	Narrative / Descriptive	1569	300
8	Narrative / Descriptive	723	650

the field for over a decade, providing the necessary data and incorporating essential handcrafted features to evaluate AES methodologies comprehensively.

Comprising an extensive repository of 12,976 essays, the ASAP dataset covers a broad spectrum of lengths ranging from 150 to 550 words. This dataset is meticulous, and the evaluation process is a distinguishing feature, ensuring the highest reliability standards. Each essay undergoes a dual-scoring procedure. To ensure robustness, each essay is assessed by a minimum of two human raters, guaranteeing a fair and comprehensive grading process. The primary objective underpinning the creation of the ASAP dataset is to encourage the exploration of cost-effective and efficient approaches for AES tasks. This objective is achieved by encompassing American students' essays from grades 7 to 10. The dataset covers eight sets, each encompassing diverse topics and genres. The initial two sets focus on argumentative essays, tasking writers with articulating their perspectives on specific subjects. Sets 3 to 6 center around response essays, challenging writers to read extracts and formulate responses based on the provided material. Finally, sets 7 and 8 delve into the domain of narrative essays, inviting writers to craft stories inspired by particular scenarios.

The ASAP dataset offers significant strengths for essay-scoring research due to its well-established presence in the academic community and comprehensive evaluation framework. Its public availability and widespread use provide a standardized benchmark. The dataset's inclusion of both holistic and analytical scoring dimensions enables a thorough assessment of essay quality, evaluating the overall effectiveness and specific components such as content, organization, and word choice.

Table 3 describes the ASAP dataset's details.

B. IMPLEMENTATION DETAILS

The custom model architecture, constructed with TensorFlow/Keras, is designed to generate essay scores. The MSE is the chosen loss function, enabling the model to fine-tune itself for accurate score predictions. K-fold cross-validation with ten folds is employed to assess its effectiveness and robustness, covering model training, validation, and rigorous testing.

Concurrently, the AdamW optimizer, a prevalent selection in DL, is employed, and a linear learning rate scheduler is integrated. Additionally, emphasis is placed on averting

TABLE 4. Hardware and software specifications.

Hardware/Software	Description
Operating System	Microsoft Windows 11 - x64-based
Physical Memory (RAM)	512 GB
GPU's	4 x NVIDIA GeForce RTX 3090
Processor	2 Intel(R) Xeon(R) Gold 6258R CPU @ 2.70GHz, 2700 Mhz, 28 Core(s), 56 Logical Processor(s)
Programming Language(s)	Python
IDE	PyCharm Professional
Framework	PyTorch

overfitting through prudent utilization of early stopping, guaranteeing effective generalization of learned patterns to new, unseen examples.

Assessing model performance involves conducting an in-depth analysis utilizing both MSE and QWK scores on a test set. The code calculates and reports the mean MSE and QWK scores across all folds.

C. COMPUTATIONAL REQUIREMENTS

Robust computational resources were essential to train and evaluate our Few-Shot essay scoring model due to the intensive demands of DL tasks involving large-scale models like BERT.

We operated on a 64-bit Microsoft Windows 11 system equipped with 512 GB of RAM, which is crucial for handling large datasets and complex models. Our setup included four NVIDIA GeForce RTX 3090 GPUs, significantly reducing training time and allowing for efficient hyperparameter tuning. The CPU setup featured two Intel Xeon Gold 6258R CPUs with 28 cores and 56 logical processors each, ensuring efficient handling of data preprocessing and model training tasks not offloaded to GPUs.

While this study utilized a high-performance environment to facilitate training, the proposed methodology is not strictly dependent on these resources for performance. Modern DL frameworks like PyTorch and TensorFlow efficiently handle scaling on various computational setups, ranging from single GPUs to even CPU-only environments. Techniques like reducing batch sizes, model distillation, and mixed-precision training enable the model to run on systems with lower specifications. This ensures that similar results can be achieved with modest hardware, although with longer training times.

We used Python as our programming language, with PyCharm Professional as our IDE for advanced code development and debugging. The PyTorch framework was chosen for its flexibility and efficiency in building and training neural networks, leveraging GPU acceleration effectively. These specifications provided a robust environment for our research. Detailed hardware and software specifications are summarized in Table 4

D. HYPERPARAMETERS SETTINGS

This experiment's hyperparameters were meticulously chosen after analyzing the training data and extensively searching

TABLE 5. Hyperparameter settings.

Hyperparameters	Values
Activation Function	GELU
Learning Rate	2e-5
Optimizer	AdamW
Batch Size	32
Training Epochs	6
No. of Folds	10
Loss Function	Mean Squared Error

for the most effective configuration. The selection process encompassed vital elements such as the optimizer, activation unit, and learning rate that demonstrated superior performance for the model. It is worth noting that hyperparameter choices can vary depending on the specific dataset, and a series of experiments were conducted to evaluate different hyperparameter settings. The combination of hyperparameters settled on consistently outperformed other configurations. The widely acclaimed GELU (Gaussian Error Linear Units) activation function, known for its excellent results and employed in the architecture of the BERT model, was adopted. The Adam optimizer was also chosen, and the batch size was set to 32, complemented by a learning rate of 2e-5 to fine-tune the model effectively. The chosen configuration aligns with recent Few-Shot learning methodologies used in natural language processing (NLP) tasks. For instance, frameworks like Pattern Exploiting Training (PET) and SetFit have demonstrated the effectiveness of Few-Shot learning in tasks such as text classification and sentiment analysis by leveraging minimal labeled examples. Similarly, GPT models, including GPT-3, have been employed with Few-Shot setups for text generation and scoring, where limited in-context examples guide the model's performance. As demonstrated in this work, these studies validate the efficacy of using a small, prompt-specific dataset to adapt pretrained models like BERT to new tasks. Table 5 describes the hyperparameter settings details.

E. FEW-SHOT SETTINGS

While the ASAP dataset contains many essays, the Few-Shot paradigm is applied at the prompt level, where only a limited number of labeled examples (support set) and queries (query set) are used for training and evaluation. This setup adheres to the principles of Few-Shot learning by leveraging pretrained models to adapt to new tasks with minimal prompt-specific labeled data.

The implementation details for AES adopted a Few-Shot learning approach with a specific configuration. In this context, the “number of shots” refers to the total essays used in a single Few-Shot task, comprising a support set (120 essays) and a query set (80 essays), totaling 200 essays per task. This configuration enables the model to adapt effectively to essay scoring by leveraging a rich yet manageable set of labeled examples for each task. The support set provides a foundation of 120 essays for the model

to learn task-specific traits. In comparison, the query set of 80 essays evaluates the model's capacity for generalization to unseen examples. This balanced configuration ensures sufficient labeled data for model adaptation while adhering to the Few-Shot learning paradigm at the prompt level, where only a limited number of examples are available for each task. This setup aligns with the principles of Few-Shot learning, emphasizing efficient use of minimal prompt-specific data rather than the entire dataset, improving the model's potential for accurate and scalable AES.

F. EVALUATION MATRIX

QWK [50] is employed as the primary metric to assess the agreement between human raters' scores and the predictions made by the model. Consistent with the conventions of the ASAP competition and previous experiments on the ASAP dataset, researchers consider QWK the gold standard for evaluation. The exclusive use of the ASAP dataset in these experiments establishes QWK as the primary criterion for evaluating the model's performance. Ranging typically between 0 (indicating no consensus among raters) and 1 (representing perfect consensus among raters), this metric offers a robust measure of the degree of agreement between humans and AES. Equation 4 outlines the calculation of the QWK score.

$$\text{QWK} = 1 - \frac{\sum_{i,j} W_{i,j} O_{i,j}}{\sum_{i,j} W_{i,j} E_{i,j}} \quad (4)$$

V. RESULTS

The subsequent section elucidates research findings and categorizes them into discrete sections. These encompass evaluating different language models without Few-Shot, outcomes with and without Few-Shot Training, and a comprehensive overview of holistic and analytical scoring results. This organized framework thoroughly examines the model's proficiency in assessing overall essay quality and its capacity to appraise specific writing attributes.

A. BERT VS. OTHER LANGUAGE MODELS

When it comes to autonomously evaluating essays, the language model used is critical to how effectively the system functions. For several reasons, BERT stands out as the ideal solution. BERT's bidirectional understanding provides a comprehensive grasp of the language by considering the context of preceding and subsequent words. This method improves its capacity to grasp compositions' general meaning and coherence. Furthermore, BERT's contextual awareness excels at capturing precise information inside the text, an essential quality for analyzing essays in which the importance of words or phrases is intimately linked to their context within the composition.

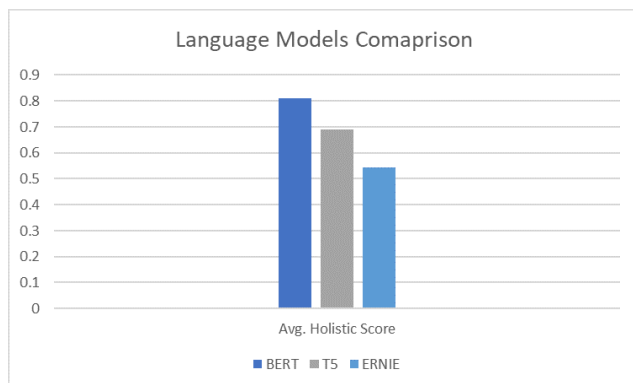
The performance gap between BERT and T5-base can be attributed to their architectural differences. BERT's bidirectional architecture provides a deeper contextual understanding of the text, which is particularly advantageous for

TABLE 6. Language model comparison.

Language Model	Avg. Holistic Score
BERT	0.812
T5-base	0.691
ERNIE	0.544

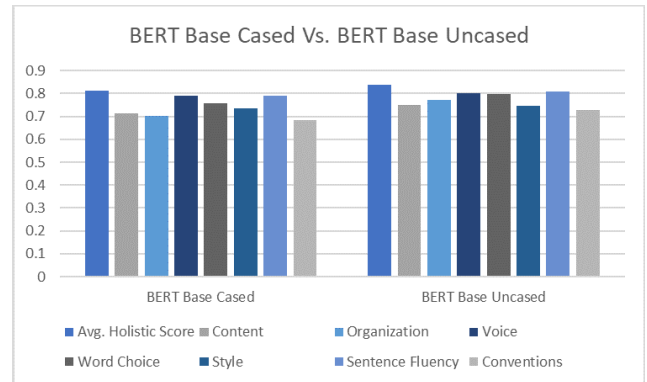
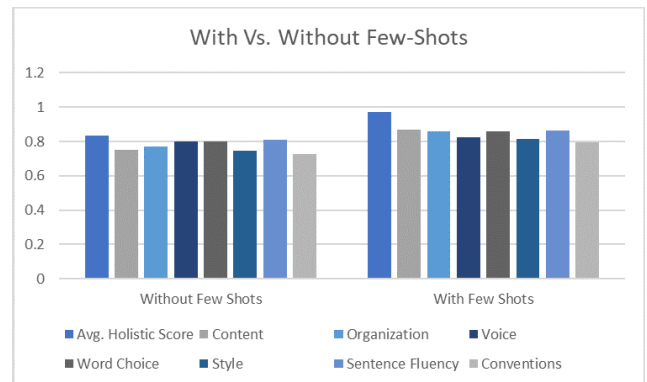
essay scoring. In contrast, T5's encoder-decoder design is more suited for generative tasks, potentially limiting its effectiveness in tasks requiring nuanced feature extraction like AES. Additionally, differences in hyperparameter tuning and fine-tuning datasets may have further contributed to the observed disparity.

A comparison was made between BERT, T5-base, and ERNIE, focusing on holistic scoring. BERT demonstrated superior performance, excelling in its ability to interpret language from both sides while considering context and effectively capturing essay details and quality. In contrast, the T5-base exhibited limitations in catching such nuances. Comparing BERT with ERNIE revealed that ERNIE, heavily trained in the Chinese language, struggled to adapt to English datasets like ASAP, resulting in poor performance. BERT's adaptability and understanding of context made it a superior choice for grading English essays. Real-world testing consistently showed BERT outperforming T5-base and ERNIE in holistic scoring, justifying its selection for the proposed AES system. Table 6 presents the comprehensive BERT, T5-base, and ERNIE results in the holistic scoring task, and Figure 8 represents the performance of the distinct models.

**FIGURE 8.** Results of distinct language models applied on asap dataset.

B. BERT BASE CASED VS. BERT BASE UNCASED

An extensive research investigation examined two prominent BERT versions: BERT Base Cased and BERT Base Uncased. The objective was to comprehensively evaluate their performance in holistic and analytical scoring tasks to gain profound insights into their capabilities. The investigation outcomes demonstrate that BERT Base Uncased consistently outshines its cased counterpart. Its consistent and superior

**FIGURE 9.** Comparative results of BERT base cased and BERT base uncased.**FIGURE 10.** Comparative results of models with and without few-shot.

performance signifies a robust tool for enhancing the accuracy and precision of such systems, promising more reliable and insightful assessments. Table 7 compares the results achieved with BERT Base Cased and BERT Base Uncased, and Figure 9 represents the performance of BERT Base Cased and BERT Base Uncased.

C. FEW-SHOT VS. WITHOUT FEW-SHOT

In this extensive study of essay scoring, a thorough examination was conducted to assess the effectiveness of integrating Few-Shot into the evaluation process. The investigation covered both holistic and analytical scoring methodologies, revealing a clear and substantial improvement in performance with the inclusion of Few-Shot. The profound impact on accuracy and reliability establishes Few-Shot as a crucial component in the AES system. Its incorporation proves its ability to bridge the gap between limited training data and comprehensive essay assessment. It highlights its potential to significantly enhance the overall quality of evaluations, promising more consistent and precise scoring. Table 8 and Figure 10 display the scoring outcomes for evaluations conducted with and without Few-Shot.

D. HOLISTIC SCORING

Researchers have explored various methodologies to enhance the accuracy and reliability of automated scoring systems in holistic essay scoring. For instance, [36] devised the

TABLE 7. Comparative results of bert base cased and bert base uncased models.

Model	Avg. Holistic Score	Content	Organization	Voice	Word Choice	Style	Sentence Fluency	Conventions
BERT Base Cased	0.812	0.713	0.701	0.792	0.757	0.736	0.792	0.684
BERT Base Uncased	0.837	0.751	0.773	0.802	0.799	0.748	0.809	0.729

TABLE 8. Comparative results of models with and without few-shot.

Type	Avg. Holistic Score	Content	Organization	Voice	Word Choice	Style	Sentence Fluency	Conventions
Without Few-Shot	0.837	0.751	0.773	0.802	0.799	0.748	0.809	0.729
With Few-Shot	0.974	0.869	0.860	0.823	0.858	0.816	0.863	0.795

Cross-prompt Trait Scorer (CTS) to tackle the challenge of limited training data by simultaneously predicting both holistic and analytical scores. Their model demonstrated an average QWK holistic score of 0.553, showcasing its ability to provide holistic assessments in diverse prompts. In the context of the PAES model, [37] presented a domain-generalized approach centered on Multi-Task Learning, reporting an average QWK holistic score of 0.536. This approach aimed to create a more adaptable essay scoring system capable of effectively handling a range of prompts.

Furthermore, [38] proposed the ProTACT model, which focused on cross-prompt essay trait scoring. Their innovative architecture incorporated prompt-specific and trait-specific encodings, enhancing the model's capability to capture nuanced essay traits across diverse prompts while retaining prompt-specific features. The ProTACT model achieved an average QWK holistic score of 0.592, emphasizing its proficiency in holistic scoring.

Reference [12] introduced a methodology that leverages transfer learning and DNN, incorporating a hierarchical fine-tuning strategy with BERT to grade essays. Their approach achieved a commendable QWK score of 0.83 when evaluated on the ASAP dataset, indicating its proficiency in assessing essay quality comprehensively.

Reference [51] introduced an innovative methodology leveraging open-source large language models (LLMs) for AES and Automated Essay Revising (AER). By employing zero-shot, Few-Shot, and p-tuning AES methods, their approach demonstrated competitive performance close to deep-learning baseline models on a human-scored essay dataset, highlighting the potential of open-source LLMs to effectively assist in essay scoring and revising tasks while maintaining data security and cost efficiency.

Reference [49] explored Few-Shot learning for AES, employing pattern-exploiting training (PET) and SetFit techniques. Their study demonstrated substantial performance gains with PET, particularly in data-scarce environments, though it is computationally intensive. SetFit, while faster, was less effective in some contexts.

Reference [52] explored the application of GPT-3.5 and GPT-4 with chain-of-thought (CoT) prompting for automatic scoring of student responses to science assessments. Their study demonstrated that CoT combined with contextual item

TABLE 9. Proposed approach vs. baseline approaches for holistic scoring.

Model	Average Holistic Score
PAES [37]	0.536
CTS [36]	0.533
BERT-MTL-Finetune [12]	0.830
ProTACT [38]	0.592
PET-SetFit [49]	-
Open-Source LLMs [51]	0.531
GPT-CoT [52]	-
BERT Base Cased	0.812
BERT Base Uncased	0.837
Proposed Model	0.974

stems and scoring rubrics significantly enhanced scoring accuracy, particularly with Few-Shot learning, highlighting the potential of advanced LLMs in educational assessment.

Unlike the baseline models, this study leverages the advanced capabilities of the BERT Base Uncased model. A comprehensive approach has been devised through meticulous hyperparameter tuning, detailed Few-Shot task construction, and robust implementation across various essay prompts and traits. This method yielded remarkable results, surpassing all previous findings with a QWK score of 0.97, indicating a substantial 15% improvement compared to prior research. This achievement underscores the effectiveness and resilience of the methodology in consistently and precisely evaluating essay quality, marking a significant stride in the field of AES. Table 9 and Figure 11 lays out the complete comparison of baseline models with the proposed model.

E. ANALYTICAL SCORING

In analytical essay scoring, raters assign scores to various dimensions assessed within the task. Researchers have diligently explored diverse methodologies to bolster the precision and dependability of automated scoring systems in this context. Notably, [36] introduced the Cross-prompt Trait Scorer (CTS) to grapple with the challenge of limited training data. Their model adeptly predicted trait-specific scores (that is analytical scores), exhibiting commendable performance across content, organization, word choice, sentence fluency, and conventions.

Within the framework of the PAES model, [37] unveiled a domain-generalized approach grounded in Multi-Task

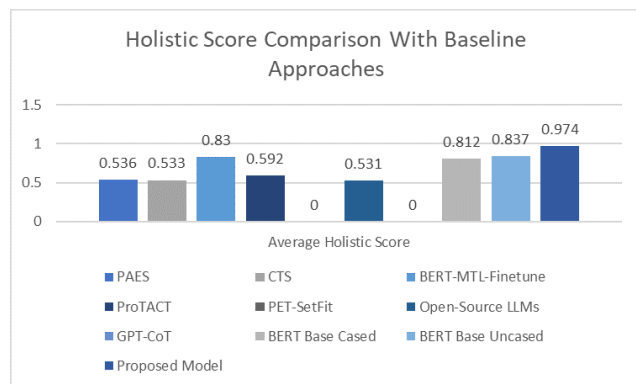


FIGURE 11. Holistic score comparison with baseline approaches.

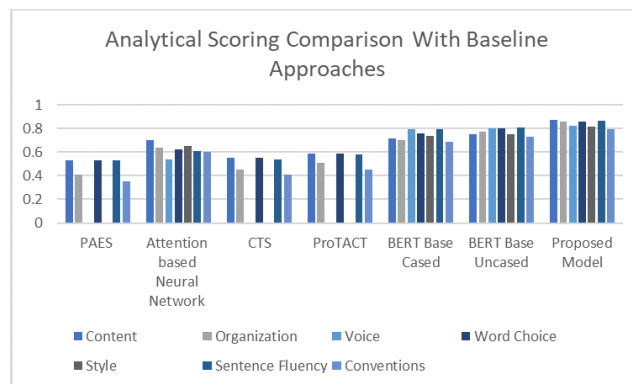


FIGURE 12. Analytical scoring comparison with baseline approaches.

Learning. This approach culminated in an average QWK score that attested to its adaptability across various traits, including content, organization, word choice, sentence fluency, and conventions.

Furthermore, [38] pioneered the ProTACT model, concentrating on cross-prompt essay trait scoring. Their innovative architectural design ingeniously incorporated both prompt-specific and trait-specific encodings. This holistic approach significantly amplified the model's capacity to discern nuanced essay traits across many prompts while faithfully retaining prompt-specific characteristics. The ProTACT model obtained high QWK scores for multiple traits, emphasizing its proficiency in analytical scoring.

Reference [32] harnessed a DNN-AES model to evaluate prompt-specific features. Employing a hierarchical representational model with an attention mechanism, they accurately predicted trait-specific essay outcomes, achieving substantial QWK scores across content, organization, style, and conventions.

In contrast to baseline models, the study leverages the intrinsic capabilities of the BERT Base Uncased model. A meticulous approach to hyperparameter tuning and dedicated implementation of Few-Shot result in remarkable improvements. Notably, the approach demonstrates significant advancements in QWK scores across traits such as content = 0.94, organization = 0.89, style = 0.81, and conventions = 0.84. This underscores the effectiveness and robustness of the methodology in consistently and precisely evaluating essay quality, marking a pioneering advancement in the field of AES. Table 10 and Figure 12 lays out the complete comparison of baseline models with the proposed model.

VI. DISCUSSION

This study aimed to enhance essay scoring by employing an analytical and holistic approach using Few-Shot transformer-based models. We ensured a standardized benchmark that promotes reproducibility and comparability by utilizing the ASAP dataset, known for its public availability and recognition within the research community. The dataset

includes both holistic and analytical scoring dimensions, enabling a comprehensive assessment of essay quality. Additionally, its diverse range of eight distinct essay prompts, covering various topics and response types, ensures our model is robust across different essay styles. The quality of annotations is crucial for the success of Few-Shot Training. The ASAP dataset, widely recognized within the research community, provided high-quality annotations, forming a reliable foundation for training our model. This high annotation quality not only contributed to the robustness of our model but also instilled confidence in its adaptability when faced with the challenge of new and unseen prompts.

Our primary objective was to evaluate the effectiveness of Few-Shot learning with pretrained BERT models in delivering accurate and reliable essay scores across various prompts and response types. The findings demonstrate the significant improvement in AES with the integration of pretrained BERT models and underscore the immense potential of Few-Shot learning in educational settings, mainly where annotated data is scarce. This promising outlook paves the way for a more efficient and reliable AES system.

A deeper analysis of the model's adaptation to different genres, such as narrative and argumentative essays, is significant. Our methodology utilizes a pretrained BERT model with Few-Shot learning, enabling effective generalization across various text types and genres. BERT's pretraining on diverse corpora equips it with a broad understanding of genre-specific linguistic patterns, allowing it to capture and adapt to the descriptive elements of narrative essays and the logical structures of argumentative ones. Few-Shot learning enhances this adaptability by enabling the model to fine-tune on minimal data from specific genres without overfitting. This flexibility ensures robust generalization, even with limited genre-specific examples. Our empirical results demonstrate the ability of the model to generalize across various prompts, including narrative and argumentative essays. This capability stems from BERT's pretrained embeddings, which capture core language patterns across genres, and the fine-tuning process that refines adaptation without requiring separate training for each genre. Overall,

TABLE 10. Proposed approach vs. baseline approaches for analytical scoring.

Model	Content	Organization	Voice	Word Choice	Style	Sentence Fluency	Conventions
PAES [37]	0.53	0.41	-	0.53	-	0.53	0.35
Attention based Neural Network [32]	0.70	0.64	0.54	0.62	0.65	0.61	0.60
CTS [36]	0.55	0.45	-	0.55	-	0.54	0.41
ProTACT [38]	0.59	0.51	-	0.59	-	0.58	0.45
BERT Base Cased	0.713	0.701	0.792	0.757	0.736	0.792	0.684
BERT Base Uncased	0.751	0.773	0.802	0.799	0.748	0.809	0.729
Proposed Model	0.869	0.860	0.823	0.858	0.816	0.863	0.795

BERT's architecture, combined with the Few-Shot learning approach, ensures adaptability across genres, which is evident in our results.

Interpretability is critical, especially in educational contexts where understanding model decisions is essential. BERT models inherently lack interpretability, but various techniques can provide valuable insights into the decision-making process. One effective technique is attention visualization. BERT's attention mechanism allows the model to weigh the importance of different words in the input text. Visualizing these attention weights provides insights into which parts of the essay the model focuses on during scoring, aligning the model's decisions with human intuition.

Few-Shot Training presents significant practical challenges, primarily due to data scarcity and the risk of overfitting. To address these issues, we utilized pretrained BERT models, which have undergone extensive pretraining on large, diverse corpora. This pretraining allows the model to learn general language patterns and contextual understanding, making it more effective even with limited task-specific data. To further mitigate the risk of overfitting, we implemented several techniques, including early stopping, k-fold cross-validation, and dropout layers. Despite the small training datasets, these methods ensured robust model performance and prevented overfitting.

Few-Shot learning can be computationally intensive due to the need for fine-tuning. Our approach optimized the training process by leveraging efficient algorithms and pretrained models, significantly reducing computational overhead while maintaining high performance. Using pretrained BERT models, trained on a vast and diverse corpus, enhanced the model's ability to generalize from a few examples, thus addressing the challenge of representing data diversity and complexity.

Our model's alignment with human evaluations is not just a theoretical concept but a tangible reality reflected in its performance, measured using the QWK score. This robust metric assesses the agreement between the models and human scores. The model's competitive results, demonstrating its reliability and effectiveness in AES, should provide reassurance about its alignment with human scoring standards. The generalizability of our BERT model, enhanced through Few-Shot Training, allows it to adapt to different essay-scoring tasks in the future. By fine-tuning on new datasets with additional traits, the model's QWK score can potentially

improve, ensuring continued alignment with human scoring standards across various contexts.

VII. CONCLUSION

This research endeavors to tackle AES by leveraging BERT's transformative capabilities and the innovative Few-Shot Training technique. The approach encompasses both holistic and analytical scoring, facilitating a comprehensive assessment of essay quality. To ensure a thorough exploration, consideration is given to BERT Base Cased and BERT Base Uncased models, both with and without Few-Shot Training, applied to the challenging ASAP dataset.

The findings reveal a clear impact of different configurations, highlighting BERT Base Uncased augmented with Few-Shot Training as the most effective combination among explored variations. BERT Base Uncased outperforms BERT Base Cased for AES, capturing a broader range of textual subtleties by treating all text in lowercase, enhancing suitability for the diverse language in essays. A Single-Task Learning model is employed for holistic scoring, while analytical scoring adopts a Multi-Task Learning approach for concurrent analytical (trait-specific) score prediction. The methodology, driven by meta-learning principles of Few-Shot Training and precise hyperparameter tuning on the BERT model, significantly outperforms baseline models, marking a substantial stride in AES.

In future work, we plan to extend our evaluation to include additional datasets with varying essay prompts and scoring criteria. This will help further demonstrate the generalizability and robustness of our model across a wider range of contexts, ensuring its effectiveness and adaptability in diverse educational settings.

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